The Representation and Visualization of Curves and Surfaces

Quantitative Engineering Analysis

1 *Overview*

Figure 1: The great thing about a boat is that it floats, no matter the level of the seas. Credit: bavariayachts.com

During the course of this module, we will be thinking about how to define and visualize the surface of a boat, and compute quantities like the displacement, center of mass, center of buoyancy, etc.

We need some building blocks about curves and surfaces in order to do this. There are very few books that are centered around curves and surfaces - these concepts typically show up in a dispersed fashion across a variety of courses, both within mathematics and in other disciplines. We will therefore be picking ideas from a variety of places, and we won't just pick ideas that are useful for designing a boat

The document you are reading will explain what you should be doing before our next class.

1.1 *Learning Goals*

By the end of this assignment, you should be comfortable with the following:

1. You should be able to use Matlab to visualize functions of one or more variables, including
   (a) In 2D and in 3D,
   (b) Explicit functions (e.g., $z = x^2 + y^2$), implicit functions (e.g., $x^2 + y^2 = 3$), and parametric functions (e.g., $x = \sin u, y = \cos u$).
2. You should be able to describe how parameters affect the curves or surfaces defined by functions.
3. You should be able to use Matlab to define and visualize curves and surfaces.
4. You should be able to generate a mathematical representation of a real physical object.

1.2 *Sources*

There is no single book or website that captures all of this material. Although we have attempted to include basic notes in this document, you will find yourself wanting more. Some good sources include:

- College Algebra. Look for sections on basic functions and their visualization.
- Single-variable and Multi-variable calculus. Look for sections on functions, curves and surfaces, parametric curves, parametric surfaces.
- The Khan Academy's material on *Thinking about Multivariable functions* is probably useful.
- Computational geometry books, 3D game rendering books, CAD books - look for sections on curves and surfaces.
- The articles developed, written, and nurtured by Eric Weisstein at Wolfram Mathworld might be a great starting point: mathworld.wolfram.com.
- Wikipedia can be a useful source, but the quality of entries on technical material is highly variable.

1.3 Suggested Approach

You are expected to do a fair amount of self-directed learning in this course. Self-directed does not mean "without help"; rather it means "with you in the driver's seat." While it's up to you how to approach this, here's what we recommend:

1. First you should quickly scan through the document, see what is being asked, and assess the extent to which you already know how to do things. Spend no more than 30 minutes or so doing this.
2. You should then read the document more closely, try out problems, and if appropriate, look at some of the other resources that are suggested. Don't spend more than a couple of hours poking around at stuff online unless it is really being productive: it's easy to spend a lot of time there without accomplishing much.
3. Then start doing the problems in earnest, and/or spend focused time with suggested resources.
4. Once you've spent a total of 4 hours working on the document, you should check your progress. Are you on track to finish within about 8 hours? Do you feel confident that you can do the stuff that's left? If not, this is when you should **ask for help.** This means **talk to a colleague**, or **talk to a ninja**, or **track down an instructor**, or **send an email to an instructor**.

2 *Getting the code*

This semester we will distribute code to you in a GitHub repository at `https://github.com/AllenDowney/QEACode`. You will start by cloning this repository from GitHub to your laptop. You can do that on the command line, like this:

```
git clone https://github.com/AllenDowney/QEACode
```

Or you can use GitHub Desktop. Either way, you should have a directory called QEACode that contains a directory called module1, which contains the MATLAB Live Script files you need for this assignment.

Later in the semester when we add more files to this repository, you will use Git to pull those changes. Note: You don't have to fork the repository on GitHub, and you don't have to set up an upstream remote.

3 *Curves - 3 hours*

Work through the MATLAB notebooks *curves1.mlx, curves2.mlx, curves3.mlx*, reading other sources as needed. When you are happy with your responses, run all cells one more time, then export the notebooks to PDF format. Submit the PDF version of the notebooks to Canvas.

4 *Surfaces - 3 hours*

Work through the MATLAB notebooks *surfaces1.mlx, surfaces2.mlx, surfaces3.mlx*, reading other sources as needed. When you are happy with your responses, run all cells one more time, then export the notebooks to PDF format. Submit the PDE version of the notebooks to Canvas.

5 *Designing Curves and Surfaces - 1 hour*

1. Pick a fruit or vegetable. Sketch it on paper from a variety of viewpoints. Now slice it in three ways, and sketch the sets of curves defined by each of these sets of slices.

2. Propose and evaluate a mathematical representation that is a good approximation to your fruit or vegetable. You could represent the entire surface, or you could design a set of curves that are good approximations to the slices.